ADITYA UNIVERSITY

Freshman Engineering Department
Department of Mathematics
Practice Questions

AY: 2025-26 **SEM-II**

Course Name & Code: Differential Equations and Vector Calculus (2501MA02).

Unit-I: First Order & First Degree Differential Equations.

Solve the following Differential Equations (From 1 to 12):

1. $\frac{dy}{dx} + \frac{y}{x \log x} = \frac{\sin 2x}{\log x}$.
2. $\frac{dy}{dx} + 2y = e^x + x, y(0) = 1.$
3. $dr + (2r \cot\theta + \sin 2\theta)d\theta = 0.$
4. $(1 + y^2)dx = (tan^{-1}y - x)dy.$
5. $(2xy + y - \tan y)dx + (x^2 - x\, tan^2 y + sec^2 y)dy = 0.$
6. $\left(y^2 e^{xy^2} + 4\, x^3\right)dx + \left(2xy\, e^{xy^2} - 3\, y^2\right)dy = 0.$
7. $\frac{dy}{dx} + \frac{y\, cos\, x + \sin y + y}{\sin x + x cos\, y + x} = 0.$
8. $(x^2y - 2\, xy^2)dx - (x^3 - 3x^2y)dy = 0.$
9. $y\,(xy \sin xy + \cos xy)dx + x(xy \sin xy - \cos xy)dy = 0$
10. $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2\, xy}$.
11. $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0.$
12. $x^4 \frac{dy}{dx} + x^3 y + cosec\,(xy) = 0.$
13. Find orthogonal trajectories of family of circles $x^2 + y^2 + 2\lambda x + c = 0$ where λ is a parameter.
14. Show that the following curves are self orthogonal curves.

    (i) $y^2 = 4a(x + a)$, a is a parameter (ii) $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$, λ is a parameter.
15. Find the orthogonal trajectories of the following curves:

    (i) $r = a(1 - \cos\theta)$, a is a parameter (ii) $r = 2a(\cos\theta + \sin\theta)$, a is a parameter.
16. Model R-L circuit, write its Differential Equation and find the current *i* at any time *t* in the circuit.
17. Find the current in the electrical circuit given by $L\frac{di}{dt} + Ri = E, i(0) = 0$ where $E = 6$ volts, $R = 100$ ohms, $L = 0.1$ henries. Also find how long will be it before the current has reached one half of its maximum value.

Unit-II: Linear Differential Equations of higher order.

Solve the following Differential Equations (From 1 to 16):

1. $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$, given that $(0) = 0, y^{|}(0) = 15$.
2. $\frac{d^4x}{dt^4} + 4x = 0.$
3. $(D^2 - 6D + 9)y = 6e^{3x} + 7e^{-2x} - log2.$
4. $(D^2 + 5D - 6)y = \sin 4x \sin x.$
5. $(D^3 + 4D)y = 5 + sin2x + 3^x.$
6. $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4.$
7. $(D^2 - 4D + 4)y = e^{2x} + 8x^2.$
8. $(D^2 - 1)y = xsin3x + \cos x$.
9. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = x^2e^{-3x}.$
10. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \sin x.$
11. $(D^2 - 4D + 4)y = 8x^2e^{2x}sin2x.$
12. $y^{||} - 6y^{|} + 9y = \frac{e^{3x}}{x^2}$ by the method of variation of parameters.
13. $\frac{d^2y}{dx^2} + a^2y = \tan ax$ by the method of variation of parameters.
14. $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = logx \sin(logx).$
15. $(2x + 3)^2\frac{d^2y}{dx^2} - (2x + 3)\frac{dy}{dx} - 12y = 6x.$
16. Solve the simultaneous equations: $x^{|} + 2y + \sin t = 0,\ y^{|} - 2x - \cos t = 0.$
17. A condenser of capacity C discharged through an inductance L and resistor R in series and the charge q at any time t satisfies the equation, $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$. If $L = 0.25$ henries, $R = 250$ ohms and $C = 2 \times 10^{-6}$ farads. Then find the charge q at any time $t > 0$, given that at initially charge $q = 0.002$ coulombs and current $\frac{dq}{dt}$ is zero in the circuit.

Unit-III: Partial Differential Equations.

Solve the following Equations (From 1 to 16):

1. $p\,tan\,x + q\,tan\,y = tan\,z.$
2. $\frac{y^2z}{x}p + xzq = y^2.$
3. $y^2p - xyq = x(z - 2y).$
4. $x^2(y - z)p + y^2(z - x)q = z^2(x - y).$
5. $px^2 - qy^2 = z(x - y).$
6. $(z^2 - 2yz - y^2)p + (xy + zx)q = (xy - zx).$
7. $(mz - ny)p + (nx - lz)q = ly - mx.$
8. $(x^2 - y^2 - z^2)p + 2xyq = 2xz.$
9. $4r + 12s + 9t = e^{3x-2y}.$
10. $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}.$
11. $(D^2 - DD^{'})z = \sin x \cos 2y.$
12. $(D^3 + D^2D^{'} - DD^{'^2} - D^{'^3})z = 3\sin(x + y).$
13. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x + y.$
14. $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2y.$
15. $(D^2 - DD^{'} - 2D^{'^2})z = (y - 1)e^x.$
16. $(D^2 + 2DD^{'} - 8D^{'^2})z = \sqrt{2x + 3y}.$

Unit-IV: Vector Differentiation.

1. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$.
2. Find the angle between the surfaces $xy = z^2$ at (4, 1, 2) and (3, 3, -3).
3. Find the values of a and b such that the surfaces $ax^2 - byz = (a + 2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1, -1, 2)$.
4. Find the directional derivative of $f(x, y, z) = xy^3 + yz^3$ at $(2, -1, 1)$ in the direction of the vector $\bar{\imath} + 2\bar{j} + 2\bar{k}$.
5. Find the directional derivative of $\emptyset(x, y, z) = xy^2 + yz^3$ at $(2, -1, 1)$ in the direction of the normal to the surface $x \log z - y^2 = -4$ at $(-1, 2, 1)$.
6. Find the directional derivative of $x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ where Q is the point $(5, 0, 4)$. Also find the magnitude of the maximum directional derivative.
7. Find the directional derivative of $xy^2 + yz^2 + zx^2$ along the tangent to the curve $x = t, y = t^2, z = t^3$ at the point $(1, 1, 1)$.
8. The temperature of points in a space is given by $T(x, y, z) = x^2 + y^2 - z$, a mosquito is located at (1,1, 2) desires to fly in such a direction that it will get warm as soon as possible. In what direction should it move. Also find what is the maximum rate of change in the temperature.
9. Find $div\ \bar{f}$ and $curl\ \bar{f}$, given that $\bar{f} = grad\ (x^3 + y^3 + z^3 - 3xyz)$.
10. Find $div\ \bar{F}$ and $curl\ \bar{F}$ at $(1, -1, 1)$, given that $\bar{F} = xy^2\,\bar{\imath} + 2x^2yz\bar{\jmath} - 3yz^2\,\bar{k}$.
11. Prove that $\text{div}(r^n\bar{r}) = (n + 3)r^n$, hence show that $\frac{\bar{r}}{r^3}$ is solenoidal.
12. Show that the vector field $\bar{F} = 2xyz^2\bar{\imath} + (x^2z^2 + z\cos yz)\bar{\jmath} + (2x^2yz + y\cos yz)\,\bar{k}$ is irrotational and find the potential function.
13. Show that the vector field $\bar{F} = (x^2 - yz)\bar{\imath} + (y^2 - zx)\bar{\jmath} + (z^2 - xy)\bar{k}$ is irrotational and hence find the scalar potential function corresponding to it.
14. Find the constants a, b, c so that the following vector $\bar{A}$ is irrotational. Also find its scalar potential function. Given that $\bar{A} = (x + 2y + az)\bar{\imath} + (bx - 3y - z)\bar{\jmath} + (4x + cy + 2z)\bar{k}$.
15. Prove that (i) $div\big(curl\ \bar{f}\big) = 0$ (ii) $curl(grad\ \emptyset) = \bar{0}$.
16. Show that $div(grad\ r^n) = \nabla^2(r^n) = n(n + 1)r^{n-2}$ where $r = |\bar{r}|$.
17. Show that $\nabla^2[f(r)] = f''(r) + \frac{2}{r}f'(r)$ where $r = |\bar{r}|$.

Unit-V: Vector Integration.

1. If $\bar{F} = 3xy\,\bar{\imath} - y^2\bar{\jmath}$, evaluate $\int_C \bar{F}\,.\overline{dr}$ where C is the curve $y = 2x^2$ in xy-plane from $(0,0)$ to $(1,2)$.
2. Find the work done in moving a particle in the force field $\bar{F} = 3x^2\,\bar{\imath} + (2xz - y)\bar{\jmath} + z\bar{k}$ along the straight line from $(0,0,0)$ to $(2,1,3)$.
3. Find the work done by the force field $\bar{F} = z\,\bar{\imath} + x\bar{\jmath} + y\bar{k}$, when it moves a particle along the arc of the curve $\bar{r} = \cos t\,\bar{\imath} + \sin t\,\bar{\jmath} - t\,\bar{k}$ from $t = 0$ to $t = 2\pi$.
4. Evaluate $\int_S \bar{F}\,.\bar{n}\,ds$ where $\bar{F} = 12x^2y\,\bar{\imath} - 3yz\bar{\jmath} + 2z\bar{k}$ and S is the portion of the plane $x + y + z = 1$ included in the first octant.
5. Evaluate $\int_S \bar{F}\,.\bar{n}\,ds$ where $\bar{F} = z\,\bar{\imath} + x\,\bar{\jmath} - 3y^2z\,\bar{k}$ and S is the surface $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.
6. Evaluate $\int_V \bar{F}dv$, given that $\bar{F} = 2xz\,\bar{\imath} - x\bar{\jmath} + y^2\,\bar{k}$ and V is the region bounded by the surfaces $x = 0, x = 2; y = 0, y = 6$ and $z = x^2,\ z = 4$.
7. Verify Green's theorem for $\int_C [(xy + y^2)dx + x^2\,dy]$, where C is bounded by $y = x$ and $y = x^2$.
8. Using Green's theorem, evaluate $\int_C [(y - \sin x)dx + \cos x\,dy]$, where C is the triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}$ and $y = \frac{2}{\pi}x$.
9. Verify Stoke's theorem for $\bar{F} = (x^2 + y^2)\bar{\imath} - 2xy\bar{\jmath}$ taken around the rectangle bounded by the lines $x = \pm a, y = 0$ and $y = b$.
10. Verify Stoke's theorem for the vector field $\bar{F} = (2x - y)\bar{\imath} - yz^2\bar{\jmath} - y^2z\bar{k}$ over the upper half of the sphere $x^2 + y^2 + z^2 = 1$ bounded by its projection on the xy-plane.
11. Verify Gauss Divergence theorem for $\bar{F} = (x^2 - yz)\bar{\imath} + (y^2 - zx)\bar{\jmath} + (z^2 - xy)\bar{k}$ taken over the rectangular parallelepiped $0 \le x \le a, 0 \le y \le b,\ 0 \le z \le c$.
12. Evaluate $\iint_S (x\,dy\,dz + y\,dz\,dx + zdx\,dy)$ by using divergence theorem , where S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.
13. State the following vector integral theorems:
    (i). Green's Theorem.
    (ii). Stokes Theorem.
    (iii). Gauss Divergence Theorem.

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